KAGGLE COMPETITION

M5 Forecasting Accuracy

Forecasting 28 Days of Walmart Unit Sales with Machine Learning

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3,049
Unique Products Tracked

28 Days
Forecast Horizon

0.59212
Best Kaggle Score

Business Context

THE CHALLENGE

Forecast 28 days of unit sales for Walmart product-store series across 10 stores in 3 U.S. states, using historical sales, calendar events, SNAP days, and price information.

WHY IT MATTERS

Inventory Efficiency

Reduce overstock and stockout risk through better demand visibility

Proactive Planning

Help teams plan four weeks ahead instead of reacting to shortages

Cost Reduction

Lower waste, holding costs, and unnecessary replenishment

Promotion & Event Readiness

Capture demand shifts from price changes, SNAP days, holidays, and local events

Data Cleaning & Preparation

INPUT → OUTPUT
3 CSVs → 1 Clean File

PIPELINE STEPS:

1. **Parse types & fill missing values**
   Dates parsed to datetime, 1,807 event NaNs filled
2. **Downcast columns**
   Sales int64 → int16, prices float64 → float32
3. **Detect leading zeros**
   77% of series have pre-shelf zeros flagged
4. **Melt wide → long & merge**
   Combined features into one row-per-item-day table
5. **Forward-fill price gaps**
   Items with late price records filled using first known price
6. **Export clean parquet**
   Single file for all downstream modeling notebooks

KEY NUMBERS:

16M+	3,049	60%
Rows After Melt	Unique Products Tracked	Of Products Had No Daily Sales

ISSUES FOUND & RESOLVED

- **1,807 event NaNs** — not missing data, just "no event"; filled with no_event variable
- **77% leading zeros —** items not yet on shelf; masked with is_active flag
- **Price gaps after melt** — items with late price records; forward/back-filled per unique product

Baseline Model Building - LightGBM

BASELINE RMSE
2.1619

WHAT IS LightGBM?

LightGBM is a gradient boosting model that designed to handle large-scale tabular data fast, and natively supports categorical features without manual encoding.

- Splits data into decision trees
- Corrects errors step by step
- Outputs sales prediction per row

FEATURE LOGIC

Categorical	Handled natively by LightGBM
Calendar	Captures weekly cycles, seasonality, and government benefit days
Price / Promo	Models consumer price sensitivity and promotional lift
Lag & Rolling	Encodes recent demand momentum

MODEL SETUP

Objective **Tweedie**	Suited for sparse count data with many zeros
Sample **2,000 products**	× 500 days
Features **14 total**	Lag, rolling means, calendar, price signals
Metric **RMSE**	Root mean squared error; lower = better
Max Rounds **2,000**	Early stopping @ 100 rounds
Learn Rate **0.05**	Found via Bayesian optimization
Num Leaves **127**	Controls tree complexity

Feature Engineering

BEST FEATURES RMSE

2.1322

METHODOLOGY

Iteratively add feature groups to the baseline model, then evaluate each on a held-out validation window using RMSE, with all lag and rolling features computed shift-first to prevent data leakage.

KEY RESULT

Three extra features (snap_flag, is_event, lag_56) beat complex combinations of other features.

FINAL FEATURE TABLE

CATEGORICAL

item_id	Item identifier
dept_id	Department ID
cat_id	Category ID
store_id	Store identifier
state_id	State identifier

CALENDAR

dow	Day of week
month	Month of year
snap_flag	SNAP benefit day
is_event	Holiday / event

PRICE / PROMO

sell_price	Item selling price
price_change	Price change vs. prev day
is_promo	Promotion flag

LAG & ROLLING

lag_7	7-day lagged sales
lag_28	28-day lagged sales
lag_56	56-day lagged sales
rmean_7	7-day rolling mean
rmean_28	28-day rolling mean

- New features added on top of baseline

Fine-Tuning: Finding the Best Model Settings

LightGBM RMSE

2.0026

Approach: 2-Phase Search Sample: 1,000 products × 500 days

Phase	Method	Trials
Phase 1	Manual Grid Search	12 sets
Phase 2	Automated Smart Search (Bayesian Optimization)	40 trials

Best Parameters Found

Parameter	Value	Meaning
Learning Rate	0.042	How fast the model learns
Tree Complexity	91 leaves	Level of detail captured
Demand Pattern	1.28	Tuned for sparse sales data
Min Data in Leaf	85	Lower = better

Model Exploration: Ridge Regression

RIDGE REGRESSION RMSE

2.1180

WHAT IS RIDGE REGRESSION?

Ridge regression is a linear model which incorporates regularization, it finds the best straight-line fit while making sure the model can still predict well with new data.

Fits a line across a graph

Shrinks noise within data

Predicts next-day sales per product

Ridge Model Setup

- Alpha = 0.5 (regularization strength)
- StandardScaler applied to all features
- Same engineered features as team models
- Same validation window for comparison

MODEL COMPARISON

Lower RMSE = better

Ridge Regression
Linear baseline
2.1180

Random Forest
Tree ensemble
2.0802

GRU
Deep learning
2.2583

Takeaways

Strength

- Fast to train, fully interpretable

Weaknesses

- Assumes linear relationships
- Produces negative predictions

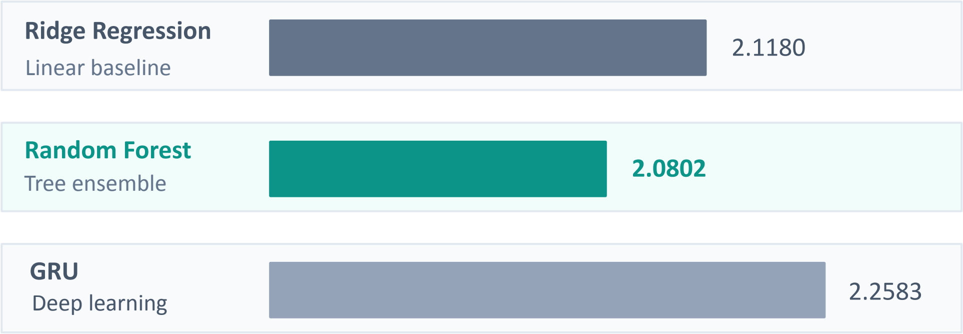
Model Exploration: Random Forest

RANDOM FOREST RMSE

2.0802

MODEL COMPARISON

Lower RMSE = better



Takeaways

Strength

- Robust to outliers and skewed data

Weaknesses

- Slower to train than Ridge or GRU
- Cannot predict beyond the training range

WHAT IS RANDOM FOREST?

Random Forest builds hundreds of decision trees on random subsets of data, then averages their predictions, can be seen as polling many experts instead of trusting one.

Grows 300 decision trees

Each tree sees random features

Averages votes for final prediction

Random Forest Model Setup

- 300 trees, max depth 20
- Min 20 samples per leaf
- Same engineered features as team models
- Same validation window for comparison

Model Exploration: GRU

GRU RMSE
2.2583

WHAT IS GRU?

A Gated Recurrent Unit (GRU) is a neural network that reads historical sales as a sequence — much like how a human analyst reviews past weeks before making a forecast.

Reads 28-day history window

Learns demand patterns over time

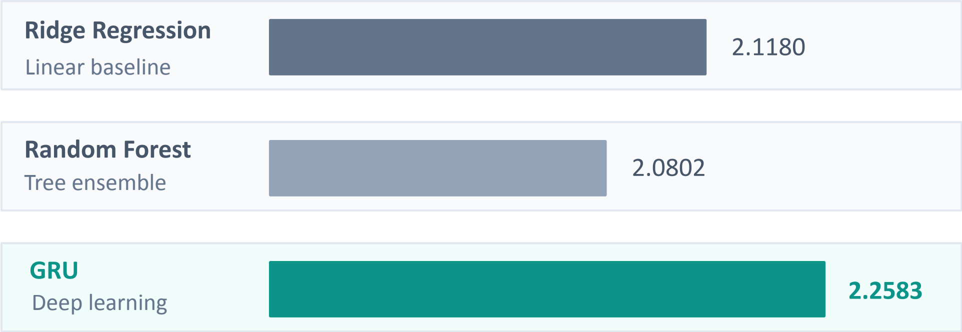
Predicts next-day sales per product

GRU Model Setup

- 28-day input window
- Same engineered features as team models
- Same validation window for comparison

MODEL COMPARISON

Lower RMSE = better



Takeaways

Strength

- Captures sequential demand patterns

Weaknesses

- Less effective on days with zero sales
- Performed the worst out of all models

Final Prediction: LightGBM

KAGGLE SCORE

0.59212

Recursive Forecasting:

1. Get static features (calendar, price, events)
2. Compute lag_7, lag_28, lag_56 from history
3. Compute rmean_7, rmean_28 from history
4. Predict demand for each ID
5. Clip negative predictions to 0

Append predictions to history → repeat

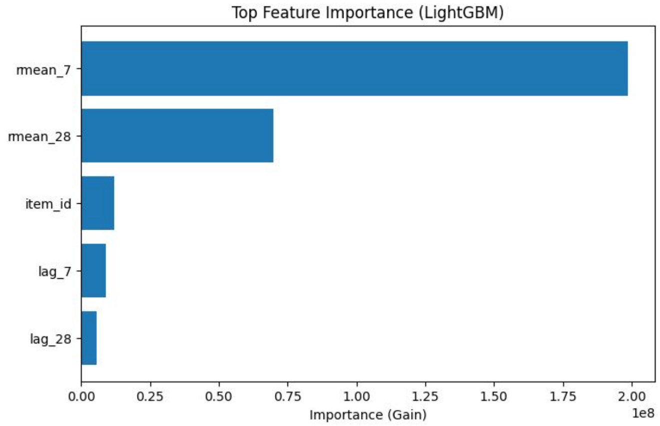
💡 **Enables multi-step prediction without retraining the model**

Ensemble Strategy:

Train multiple models with three different seeds

👉 Reduces variance & Improves stability

📈 **~0.001 WRMSSE improvement vs single model**



The model learns both temporal dynamics and item-level behavior

Recommendations

Business Impact

- Improves visibility into future demand of 3,049 products sold across 10 stores
- Helps inventory teams plan and allocate stock earlier
- Reduces waste from over- or under-preparation

Risks & Limitations

- Sudden demand shifts (e.g. supply disruptions) may reduce accuracy
- Promotions or local events may need manual adjustment
- Forecast uncertainty increases further out in the 28-day horizon

Next Steps

- Refresh the model periodically with new sales data
- Incorporate additional business context (e.g. pricing, events) into future forecasts
- Use forecast output alongside manager judgment — not as a replacement